

Confidence Intervals for Population Mean

Statistics Worksheet · Grade 11–12 / Introductory College Statistics

Name: _____

Date: _____

Learning Objectives

- Identify and organize given values (sample size, sample mean, population standard deviation, confidence level) from word problems
- Verify conditions for constructing a valid confidence interval for a population mean
- Calculate the critical z-value, margin of error, and confidence interval using the formula $\bar{x} \pm z^* \cdot (\sigma / \sqrt{n})$

Problems

1. A simple random sample of 100 students was taken from a normally distributed population. The sample mean score on a reading test was 520 points, and the population standard deviation is 80 points. Identify the values of n , $\bar{x}$, σ , and the confidence level if you are constructing a 95% confidence interval.

$$n = ?, \quad \bar{x} = ?, \quad \sigma = ?, \quad C = ?$$

2. To construct a confidence interval for a population mean, two conditions must be verified. State both conditions and check whether they are satisfied for this scenario: An SRS of 200 adults was drawn from a normally distributed population.

3. Find the critical z-value (z-star) for a 90% confidence level. Use the formula (1 minus C) divided by 2 to find the tail area, then use the standard normal table or calculator.

$$z^* \text{ for } C = 0.90$$

4. Find the critical z-value (z-star) for a 99% confidence level.

$$z^* \text{ for } C = 0.99$$

5. A normally distributed population has a standard deviation of 50. An SRS of 400 people is taken with a sample mean of 300. Calculate the margin of error for a 95% confidence interval. Use z-star = 1.96.

$$E = z^* \cdot \frac{\sigma}{\sqrt{n}}$$



6. Using the information from the video: an SRS of 500 students from a normally distributed population has a sample mean SAT math score of 461, and the population standard deviation is 100. Construct the 95% confidence interval for the population mean. Use z-star = 1.96.

$$\bar{x} \pm z^* \cdot \frac{\sigma}{\sqrt{n}}$$

7. A researcher takes an SRS of 64 employees from a normally distributed population. The sample mean salary is \$45,000 and the population standard deviation is \$8,000. Construct a 90% confidence interval for the true population mean salary. Use z-star = 1.645.

$$\bar{x} \pm z^* \cdot \frac{\sigma}{\sqrt{n}}$$

8. A 95% confidence interval for a population mean is reported as (112.4, 127.6). Find the sample mean and the margin of error from this interval.

$$\bar{x} = \frac{\text{Lower} + \text{Upper}}{2}, \quad E = \frac{\text{Upper} - \text{Lower}}{2}$$

9. A health researcher wants to estimate the mean cholesterol level of adults in a city. From a normally distributed population with a known standard deviation of 35 mg/dL, an SRS of 225 adults yields a sample mean of 198 mg/dL. Construct a 99% confidence interval and write a concluding statement. Use z-star = 2.576.

$$\bar{x} \pm z^* \cdot \frac{\sigma}{\sqrt{n}}$$

10. Compare the widths of the 90%, 95%, and 99% confidence intervals for a population with sigma = 120, constructed from an SRS of 900. The sample mean is 850. Calculate all three intervals and explain how the confidence level affects the width of the interval.

$$\bar{x} \pm z^* \cdot \frac{\sigma}{\sqrt{n}}, \quad z_{90}^* = 1.645, \quad z_{95}^* = 1.96, \quad z_{99}^* = 2.576$$



Confidence Intervals for Population Mean — Answer Key

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Answer Key

1. Answer: $n = 100$, $\bar{x} = 520$, $\sigma = 80$, $C = 0.95$

- Read the problem: sample size = 100, so $n = 100$.
- The sample mean is stated as 520, so $\bar{x} = 520$.
- The population standard deviation is given as 80, so $\sigma = 80$.
- The confidence level is 95%, written as $C = 0.95$.

2. Answer: Condition 1: Sample is randomly selected — satisfied (SRS). Condition 2: Population is normally distributed — satisfied (stated in problem).

- Condition 1: The sample must be randomly selected. The problem states it is an SRS, so this condition is satisfied.
- Condition 2: The population must be normally distributed (or the sample size large enough for CLT). The problem states the population is normally distributed, so this condition is satisfied.
- Both conditions are met; it is valid to proceed with the confidence interval.

3. Answer: $z^* = 1.645$

- Tail area = $(1 - 0.90) / 2 = 0.10 / 2 = 0.05$.
- We need the z-value where the cumulative area to the left is $1 - 0.05 = 0.95$.
- From the standard normal table or calculator (inverse normal): $z^* = 1.645$.

4. Answer: $z^* = 2.576$

- Tail area = $(1 - 0.99) / 2 = 0.01 / 2 = 0.005$.
- We need the z-value where the cumulative area is $1 - 0.005 = 0.995$.
- From the standard normal table or inverse normal on a calculator: $z^* = 2.576$.

5. Answer: $E = 4.9$

- Identify values: $z^* = 1.96$, $\sigma = 50$, $n = 400$.
- Standard error = $\sigma / \sqrt{n} = 50 / \sqrt{400} = 50 / 20 = 2.5$.
- Margin of error $E = 1.96 \times 2.5 = 4.9$.

6. Answer: (452.23, 469.77)

- Given: $\bar{x} = 461$, $z^* = 1.96$, $\sigma = 100$, $n = 500$.
- Standard error = $100 / \sqrt{500} = 100 / 22.36 \approx 4.47$.
- Margin of error $E = 1.96 \times 4.47 \approx 8.77$.
- Lower bound = $461 - 8.77 = 452.23$.
- Upper bound = $461 + 8.77 = 469.77$.
- The 95% confidence interval is (452.23, 469.77).



7. Answer: (\$43,355, \$46,645)

- Given: $\bar{x} = 45000$, $z^* = 1.645$, $\sigma = 8000$, $n = 64$.
- Standard error = $8000 / \sqrt{64} = 8000 / 8 = 1000$.
- Margin of error $E = 1.645 \times 1000 = 1645$.
- Lower bound = $45000 - 1645 = 43355$.
- Upper bound = $45000 + 1645 = 46645$.
- The 90% confidence interval is (\$43,355, \$46,645).

8. Answer: $\bar{x} = 120$, $E = 7.6$

- Sample mean = $(112.4 + 127.6) / 2 = 240 / 2 = 120$.
- Margin of error $E = (127.6 - 112.4) / 2 = 15.2 / 2 = 7.6$.
- Interpretation: The confidence interval is centered at 120 with a margin of error of 7.6.

9. Answer: (192.0, 204.0); We are 99% confident the true mean cholesterol level lies between 192.0 and 204.0 mg/dL.

- Given: $\bar{x} = 198$, $z^* = 2.576$, $\sigma = 35$, $n = 225$.
- Verify conditions: SRS (yes) and normally distributed population (yes).
- Standard error = $35 / \sqrt{225} = 35 / 15 \approx 2.333$.
- Margin of error $E = 2.576 \times 2.333 \approx 6.01$.
- Lower bound = $198 - 6.01 = 191.99 \approx 192.0$.
- Upper bound = $198 + 6.01 = 204.01 \approx 204.0$.
- Conclusion: We are 99% confident the true mean cholesterol level lies between 192.0 and 204.0 mg/dL.

10. Answer: 90%: (843.42, 856.58); 95%: (842.16, 857.84); 99%: (839.71, 860.29). Higher confidence level → wider interval.

- Given: $\bar{x} = 850$, $\sigma = 120$, $n = 900$. Standard error = $120 / \sqrt{900} = 120 / 30 = 4$.
- 90% CI: $E = 1.645 \times 4 = 6.58$. Interval: $(850 - 6.58, 850 + 6.58) = (843.42, 856.58)$. Width = 13.16.
- 95% CI: $E = 1.96 \times 4 = 7.84$. Interval: $(850 - 7.84, 850 + 7.84) = (842.16, 857.84)$. Width = 15.68.
- 99% CI: $E = 2.576 \times 4 = 10.30$. Interval: $(850 - 10.30, 850 + 10.30) = (839.70, 860.30)$. Width = 20.60.
- Conclusion: As the confidence level increases, the critical z-value increases, resulting in a larger margin of error and a wider confidence interval. Higher confidence comes at the cost of less precision.

